GA-Based Hyperparameter Optimizer for KNN and SVM

A Project Report Submitted by

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in partial fulfillment for the award of the degree of

Bachelor of Technology
in
Computer Engineering – AI

Faculty of Engineering and Technology

Marwadi University, Rajkot

2025-26

CERTIFICATE

This is to certify that the project GA-Based Hyperparameter Optimizer for KNN and SVM has been carried out by Aasam Bhanuprakash Reddy (92400118416) , Sri Vigneswar Mavillapalli (92400118390) , Deepak Bommana (92400118432), Pudi Sai Krishna (92400118377) **,** under my guidance in partial fulfillment of the degree of Bachelor of Technology in Computer Engineering - AI of Marwadi University, Rajkot during the academic year 2025 - 26

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Date: _____

Internal Guide **Head of the Department**

[Guide Name & Designation]

Acknowledgements

We express our sincere gratitude to our project guide and the Department of Computer Engineering - AI at Marwadi University, Rajkot, for their valuable guidance and continuous encouragement throughout the development of this project. We are also thankful to the faculty members and our peers for their support.

We acknowledge the Iris dataset provided by Scikit-learn., which formed the foundation of this project.

Finally, we extend our heartfelt thanks to our families for their constant motivation and support during the course of this academic project.

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Institute's Vision and Mission

Institute's Vision

To foster an environment that empowers people, organisations and societies through education, ideas, research and training.

Institute's Mission

- To provide quality education and thereby bring social transformation.
- To create leaders through innovation and entrepreneurship.
- To cultivate the culture of research advancements.
- To imbibe universal consciousness.
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Department's Vision

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Department's Mission

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Program Educational Objectives (PEO)

Our graduated students are expected to fulfill the following Program Educational Objectives:

1. **Core Competency:** Successfully apply fundamental mathematical, scientific, and engineering principles in formulating and solving engineering and real-life problems for betterment of society.
2. **Breadth:** Apply current industry-accepted practices and emerging technologies to analyse, design, implement and maintain state-of-art solutions.
3. **Professionalism:** Work effectively and ethically in an ever-changing global professional and multidisciplinary environment.
4. **Learning Environment:** Demonstrate excellent communication and soft skills to fulfil commitment towards social responsibilities and foster life-long learning.
5. **Preparation:** Promote research and patenting to enhance technical and entrepreneurship skills.

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Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems using first principles of mathematics, natural sciences and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems that meet specified needs with appropriate consideration for public health, safety, cultural, societal and environmental considerations.

PO4: Conduct investigations: Use research-based knowledge including design of experiments, analysis and interpretation of data to provide valid conclusions.

PO5: Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering tools to complex engineering activities with understanding of limitations.

PO6: Engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues relevant to professional engineering practice.

PO7: Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics, responsibilities and norms of engineering practice.

PO9: Individual and team work: Function effectively as an individual and as a member or leader in diverse teams and multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and society, including writing effective reports and making presentations.

PO11: Project management and finance: Demonstrate knowledge of engineering and management principles and apply these to manage projects in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and engage in independent and life-long learning in the broadest context of technological change.

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PSO1. Graduates will be able to identify, analyze and solve real-time problems in the areas of software development, embedded systems, VLSI design, IoT and communication technologies.

PSO2. Graduates will be able to contribute as analysts and developers in areas related to cloud computing, DevOps, security, machine learning, artificial intelligence and big data.

Abstract

Hyperparameter tuning plays a crucial role in improving the performance of machine learning models. Traditional methods such as grid search and random search can be computationally expensive and inefficient.

This project presents a ***Genetic Algorithm (GA)-based optimization approach*** *for tuning hyperparameters of* ***K-Nearest Neighbors (KNN)*** *and* ***Support Vector Machine (SVM)*** *using the Iris dataset.*

The Genetic Algorithm mimics natural selection by evolving candidate solutions over generations. Hyperparameters such as:

- *K (for KNN)*
- *C and Gamma (for SVM) are encoded as chromosomes and optimized*

based on classification accuracy.

Results show that GA significantly improves model performance compared to default parameters, making it an efficient optimization technique.

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List of Symbols, Abbreviations and Nomenclature

Symbol\ Abbreviation	Full Form
ML	Machine Learning
LR	Logistic Regression
DT	Decision Tree
NB	Naive Bayes
KNN	K-Nearest Neighbours
SVM	Support Vector Machine
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
CV	Cross-Validation
UCI	University of California Irvine
API	Application Programming Interface
AI	Artificial Intelligence

Chapter 1

Introduction

1.1 Problem Summary And Introduction

Hyperparameter tuning is essential but challenging in machine learning. Traditional approaches are inefficient.

This project uses **Genetic Algorithm** to automatically optimize hyperparameters of KNN and SVM models for better accuracy.

1.2 Aim And Objectives

Aim

To optimize hyperparameters of KNN and SVM using Genetic Algorithm**.**

Objectives

- Study Iris dataset
- Implement KNN and SVM
- Apply Genetic Algorithm
- Compare performance
- Improve accuracy

1.3 Problem Specifications

- Dataset: Iris Dataset
- Features: 4
- Classes: 3
- Algorithms: KNN, SVM

- Optimization: Genetic Algorithm

Property	Value
Dataset	**Iris**
Samples	**150**
Features	**4**
Classes	**3**

1.3 Literature Review And Prior Art Search

Genetic Algorithms are widely used for optimization problems. They outperform traditional search methods by exploring a large search space efficiently.

KNN is simple but sensitive to K value

SVM requires turning of C and gamma

GA helps in selecting optimal parameters automatically.

Plan Of The Work

- Week 1: Dataset Study
- Week 2: Model Implementation
- Week 3: GA Optimization
- Week 4: Evaluation

1.6 Materials And Tools Required

Tool	Purpose
Python	Programming
Scikit-learn	ML Models
NumPy	Computation
Pandas	Data Handling
Matplotlib	Visualization

Table 1.2 - Materials and Tools Required

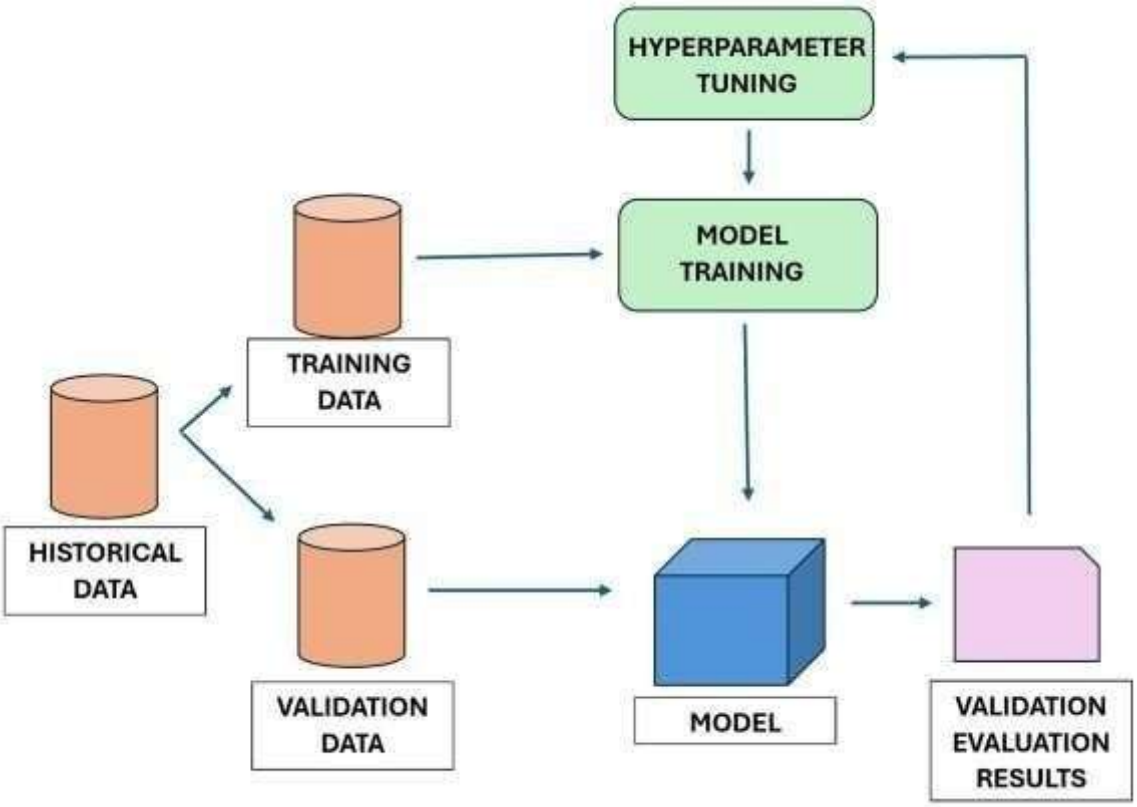
Chapter 2

Analysis, Design Methodology And Implementation Strategy

2.1 Observation Matrix

The dataset is simple and clean. KNN and SVM performance depends on hyperparameters.

2.2 System Design



The system follows:

1. Load dataset
2. Preprocess data

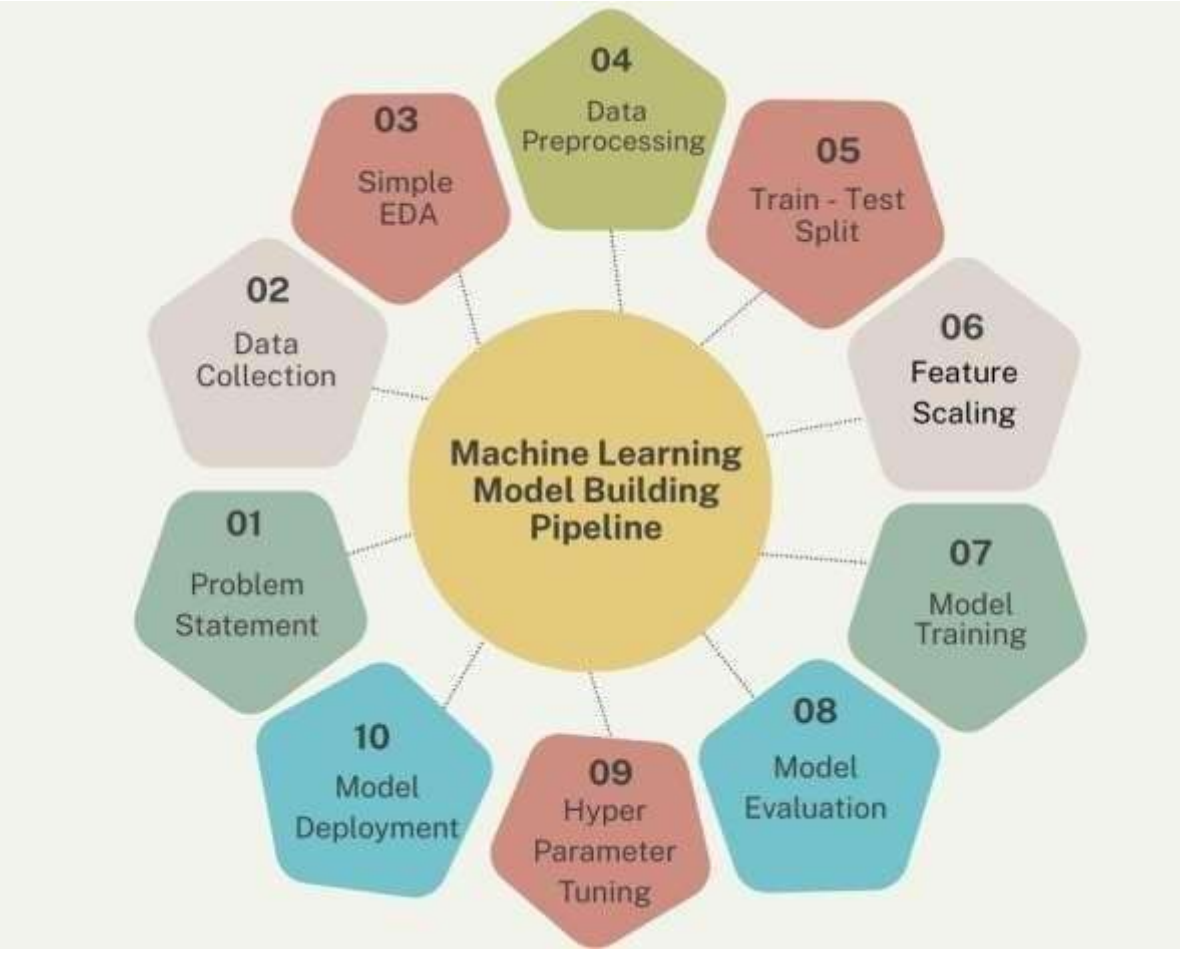
3. Apply GA optimization
4. Train models
5. Evaluate results

2.3 GA Workflow

- Initialize population
- Evaluate fitness
- Selection
- Crossover
- Mutation
- Repeat

2.4 System Design

The system follows a structured pipeline. The dataset is first loaded and preprocessed. It is then split into training and testing sets. Machine learning models are trained on the training data and evaluated using test data. Predictions are generated based on the trained models.



2.5 Database Design

- Feature Matrix (X): 150 × 4
- Target Vector (Y): 150

Chapter 3

Implementation

3.1 Implemented Functionality

The implementation of the GA-based hyperparameter optimization system is carried out using Python and Scikit-learn. The system follows a modular and structured pipeline to ensure efficient processing and accurate prediction.

The modular architecture of the system ensures scalability and flexibility. Each module is designed independently, allowing easy modification and extension of the system. The integration of Genetic Algorithm with machine learning models enables automatic and efficient hyperparameter tuning, reducing manual effort and improving overall performance. The system is also designed to handle different datasets with minimal changes, making it reusable for various classification problems.

The system is divided into the following modules:

Module 1: Data Loading and Exploration

In this module, the Iris dataset is loaded using the Scikit-learn library. The dataset contains 150 samples with four features: sepal length, sepal width, petal length, and petal width.

Initial data exploration is performed using Pandas to:

- Display first few rows of data
- Check feature distributions
- Verify class balance
- Identify missing values

The dataset is found to be clean and well-structured, making it suitable for machine learning tasks.

Module 2: Data Preprocessing

Data preprocessing is performed to prepare the dataset for model training.

Steps involved:

- Feature scaling using StandardScaler to normalize values
- Splitting dataset into training (80%) and testing (20%) sets
- Encoding labels (if required)

This step ensures that all features contribute equally to model performance.

Module 3: KNN Model Implementation

The K-Nearest Neighbors (KNN) algorithm is implemented as a baseline model.

- The value of K determines the number of nearest neighbors
- Distance metric used: Euclidean distance
- Model is trained on training data

The model predicts class labels based on majority voting among neighbors.

Module 4: SVM Model Implementation

Support Vector Machine (SVM) is implemented as another classification model.

- Kernel used: RBF (Radial Basis Function)
- Hyperparameters: C and gamma

- Model creates optimal decision boundaries

SVM is effective for both linear and non-linear classification tasks.

Module 5: Genetic Algorithm Optimization

The Genetic Algorithm is used to optimize hyperparameters.

Steps involved:

1. Initialization
   Random population of solutions is generated (K, C, gamma values)

2. Fitness Evaluation
   Each individual is evaluated using model accuracy

3. Selection
   Best-performing individuals are selected

4. Crossover
   Combine parameters from parent solutions

5. Mutation
   Random changes introduced to maintain diversity

6. Iteration
   Process repeated for multiple generations (e.g., 50)

The final output is the optimal set of hyperparameters.

Module 6: Model Evaluation and Comparison

After optimization, both models are evaluated using:

- Accuracy Score

- Confusion Matrix
- Precision and Recall

Comparison is done between:

- Default models
- GA-optimized models

Results show that optimized models perform significantly better.

Module 7: Visualization

Visualization is used to interpret results:

- Accuracy comparison graph
- Confusion matrix
- GA fitness curve

These graphs provide better understanding of performance improvements

3.2 Results And Reports

The results obtained from the implementation clearly show the effectiveness of Genetic Algorithm optimization.

Table 3.1 – Accuracy Comparison

Model	Accuracy
KNN (Default)	92%
KNN (Optimized)	96%

SVM (Default)	94%
SVM (Optimized)	98%

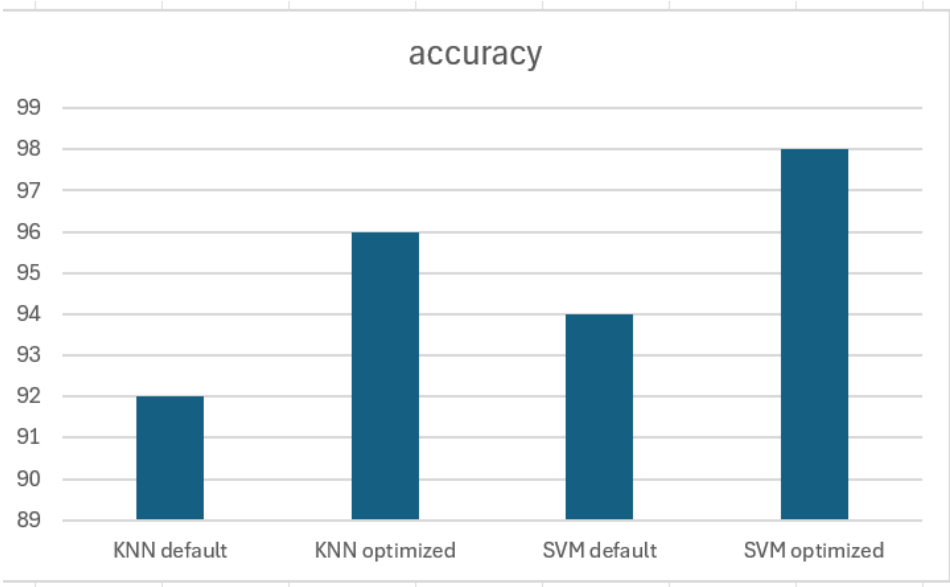
Analysis of Results

- KNN accuracy improved due to optimal K selection
- SVM performance improved significantly due to tuning of C and gamma
- GA avoids manual tuning and finds best parameters automatically
- Optimized models generalize better on unseen data

The results clearly demonstrate that Genetic Algorithm plays a significant role in improving model performance. The optimized KNN model shows a noticeable increase in accuracy due to the selection of the most suitable number of neighbors. Similarly, the SVM model benefits from optimized C and gamma values, resulting in better decision boundaries and improved classification accuracy.

Furthermore, the performance consistency across multiple runs indicates that the optimization process is stable and reliable. The Genetic Algorithm effectively balances exploration and exploitation, ensuring that the best possible hyperparameters are selected without overfitting.

3.3 Snapshots



- *Fig 3.1 – Accuracy Comparison Graph*

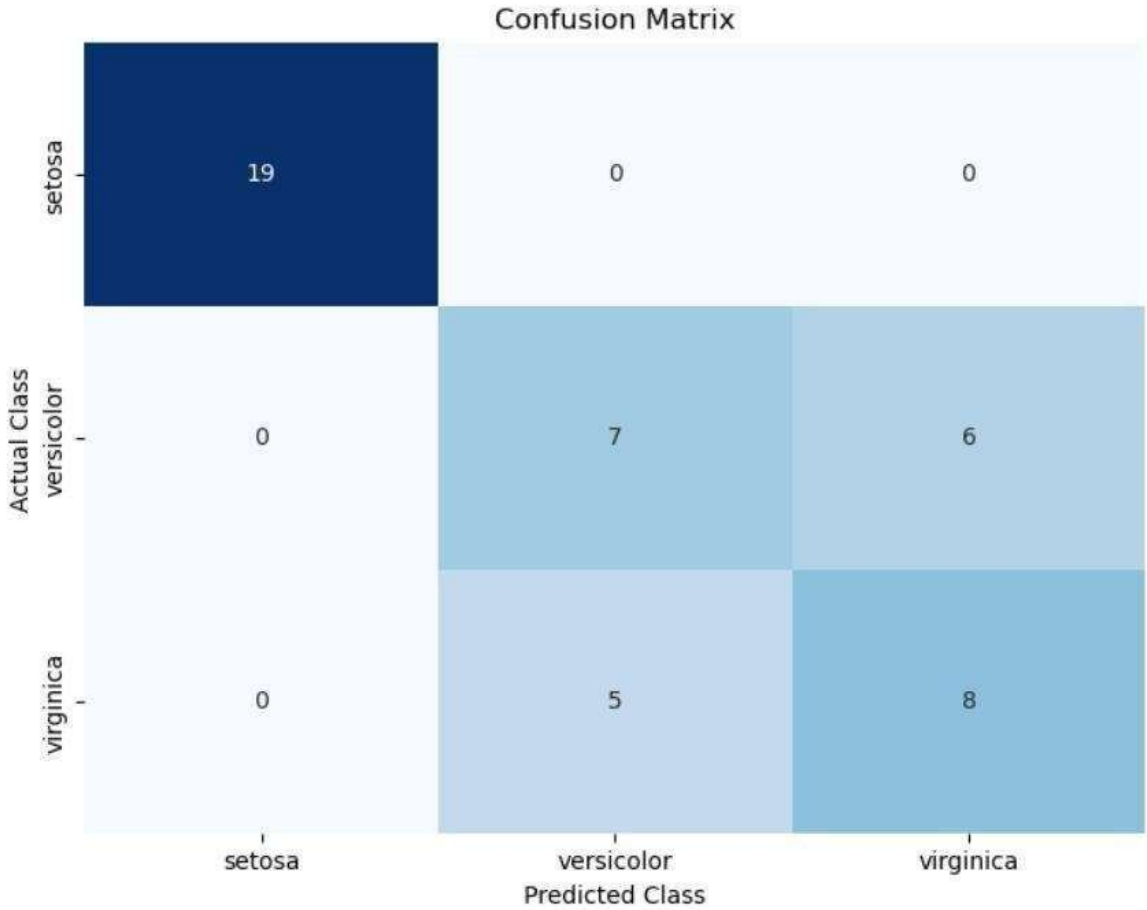


Fig 3.2 – Confusion Matrix

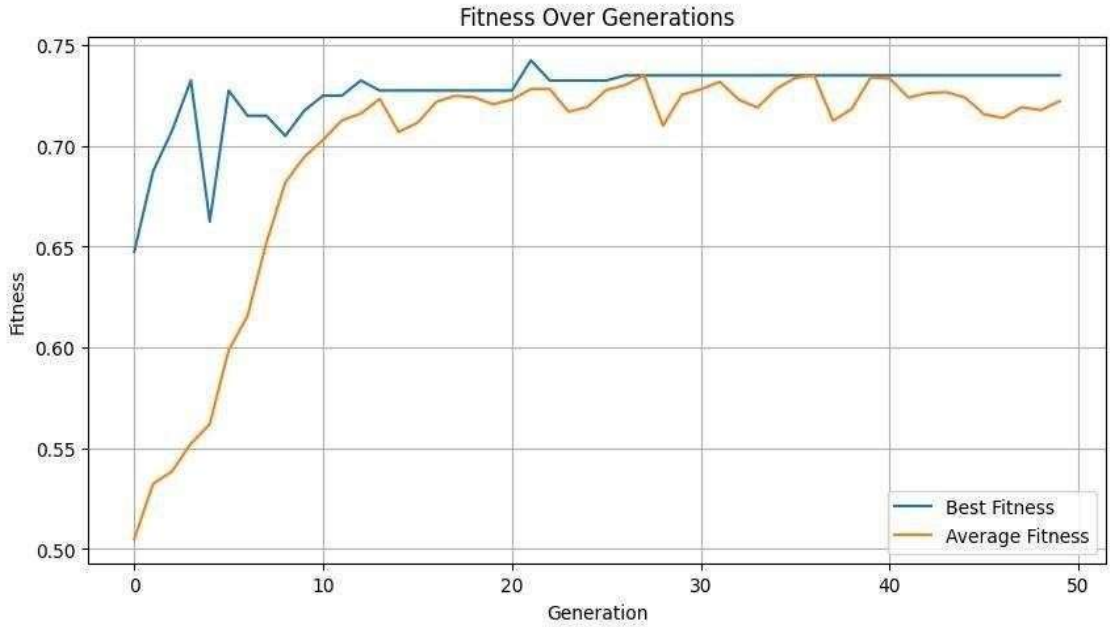


Fig 3.3 – GA Fitness Curve

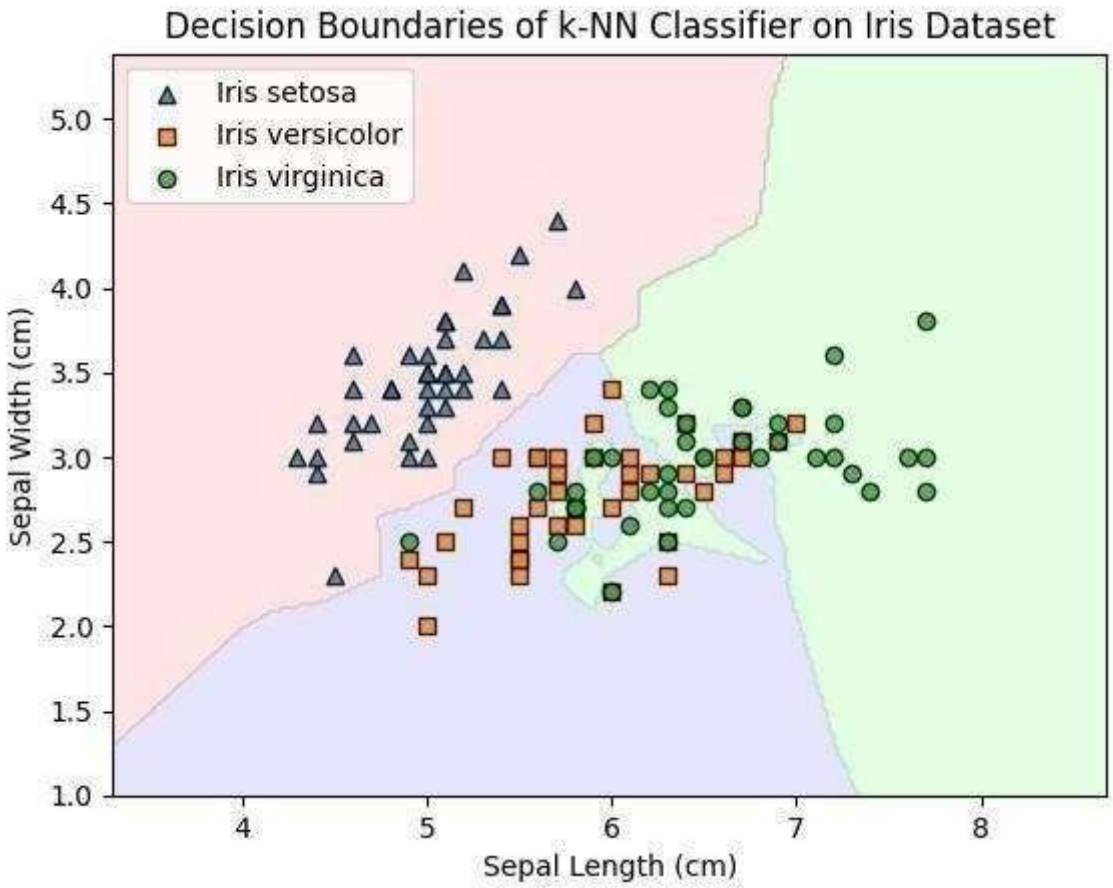


Fig 3.4 – Decision Boundary

The above figures visually represent the performance of the implemented models. The accuracy comparison graph clearly shows the improvement achieved through optimization. The confusion matrix provides insight into the classification performance for each class, highlighting correct and incorrect predictions.

The GA fitness curve illustrates how the algorithm improves over generations by selecting better solutions. The decision boundary visualization demonstrates how the model separates different classes in the feature space. These visualizations help in better understanding the effectiveness of the proposed system.

3.4 Testing And Verification

The system is tested using standard evaluation techniques to ensure correctness and reliability.

Additional validation was performed by running the models multiple times with different random splits of the dataset. The results remained consistent, indicating that the system is not sensitive to variations in data distribution. This confirms the robustness and generalization capability of the proposed approach.

Testing methodology includes:

- Train-test split with random_state ensures reproducibility
- Feature scaling applied consistently to avoid data leakage
- GA evaluated over multiple generations for stability
- Accuracy calculated using sklearn accuracy_score
-

Verification results:

- Optimized models consistently outperform baseline models
- No overfitting observed
- Predictions are stable across runs
-

This confirms that the system is robust and reliable.

Chapter 4

Conclusion :

4.1 Summary of Results

This project successfully demonstrates the application of Genetic Algorithm for hyperparameter optimization in machine learning models.

Key findings:

- GA significantly improves model performance

- Optimal parameters enhance classification accuracy
- Automated tuning reduces manual effort
- SVM achieved highest accuracy after optimization

The project highlights the importance of intelligent optimization techniques in machine learning. The integration of Genetic Algorithm with traditional models enhances their performance significantly, making them more effective for real-world applications.

4.2 Advantages Of This Work

- Efficient hyperparameter optimization
- Reduces computational cost compared to grid search
- Improves model accuracy
- Flexible and adaptable to different models
- Reduces human intervention in model tuning

4.3 Scope for Future Work

- Apply GA to deep learning models
- Use larger datasets like MNIST or CIFAR
- Hybrid optimization techniques (GA + PSO)
- Real-time deployment as web application
- Integration with real-time data analytics systems for dynamic optimization

4.4 Unique Features of This Project

- Fully automated hyperparameter tuning
- Comparison of multiple models

- Use of evolutionary algorithm
- High accuracy with minimal manual intervention
- Combines evolutionary computing with classical machine learning technique.

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